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Joveo Powers AI-Driven Prediction and Auto-Trading Engine with Microservices Architecture on AWS

Joveo, an AI-powered, fully integrated talent sourcing platform, transitioned from a monolithic architecture to microservices architecture on AWS to support scale and focus on innovation. As a result, the business introduced new offerings that help its customers hire more qualified applicants, cut costs, and improve application and hire conversion rates.

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33%

more qualified job applicants and hires for clients

25%

reduction in costs per hire delivered



2

350 TB

of data processed and stored on Amazon S3

Overview

[Joveo](#) provides an end-to-end, fully integrated talent sourcing solution that uses machine learning to optimize recruitment marketing campaigns. To support growth and reduce infrastructure management time, Joveo transitioned from a monolithic architecture to a microservices architecture on AWS.

Joveo uses AWS services such as [AWS Batch](#), [Amazon Simple Storage Service](#) (Amazon S3) and [Amazon Elastic Kubernetes Service](#) (Amazon EKS) to increase scalability and foster a culture of innovation for its developers. As a result, Joveo has introduced products that help customers source 33 percent more qualified job applicants, reduce costs by over 25 percent, and improve click-to-application and application-to-hire conversion rates.



Opportunity | Scaling and Innovating to Transform J with AI-Driven Solutions



In 2017, Joveo embarked on an ambitious journey to deliver the right job to the right person anywhere in the world, using data science and machine learning. Today, Joveo is closer than ever to realizing this vision. The company offers comprehensive talent sourcing solutions for employers, staffing firms, recruitment process outsourcing (RPO), and recruitment marketing agencies. Machine Optimized Jobs (MOJO), the company's artificial intelligence (AI)-driven programmatic job advertising platform, harnesses data science and machine learning to consistently match job placements with the most relevant talent sources and audiences.

Joveo has been among America's fastest-growing private companies for the past three years, landing on the annual Inc. 5000 list. Joveo's engineers were looking at various approaches to scale to meet this demand. Prateek Mishra, vice president of engineering at Joveo, says, "We adopted a monolithic architecture on Amazon Web Services [AWS], comprising a few machines and databases. However, we faced time-consuming infrastructure management tasks since the architecture couldn't support our expansion and a higher volume of job placements."

Joveo wanted to reduce the time developers spent on system engineering and writing sophisticated code to schedule processing jobs. The business sought to offload these tasks to give its developers more time to create innovative products, such as a real-time media planning model.

Solution | Streamlining Infrastructure Management and Processing on AWS

Joveo's MOJO platform runs on AWS, utilizing services such as [Amazon Elastic Compute Cloud](#) (Amazon EC2) and Amazon S3, which the company uses to process and store large numbers of jobs.

To run computational jobs at scale without managing compute resources, the company replaced its internal batch processing framework with AWS Batch, a fully managed batch computing service that schedules and runs containerized batch workloads. More recently, Joveo adopted a microservices architecture and moved to Amazon EKS, a managed Kubernetes service to run Kubernetes on AWS.

Building on AWS, particularly AWS Batch, Joveo can seamlessly scale MOJO and its developers can focus on building new products instead of managing and maintaining an internal batch processing solution. Joveo's processes and stores close to 350 TB of data in Amazon S3.



Mishra says, "By using AWS Batch and Amazon EKS, we eliminate concerns about infrastructure management and hardware provisioning to support scale. This frees up our developers to fully focus on innovation and the development of new products."

Since migrating to a microservices architecture on AWS, Joveo expanded its solution offerings to include an AI-driven prediction and auto-trading engine that automatically suggests the best job postings, publishers, bids, and budget allocation based on customers' requirements and executes job advertising campaigns to deliver the best results.

Outcome | Developing Innovative Products for Better Insight into Ad Performance

As Joveo continues to expand its use of AWS services, the company also introduced new offerings such as career site building capabilities, a content management system purpose-built for recruitment, candidate relationship management (CRM), and a candidate engagement platform, to encompass all recruitment marketing needs of high-volume employers.

By integrating MOJO with its new product offerings, Joveo has significantly contributed to its customers' success; many customers have seen a boost of over 33 percent in qualified job applicants alongside a reduction of more than 25 percent in recruitment marketing costs.

Mishra states, "By running our platforms on AWS, we can help our customers gain a comprehensive understanding of job ad performance across different locations and categories. Specifically, our prediction engine and auto-trader use this data to post jobs to the most effective publishers, where they're likely to see the best results. We provide a rich data experience to our customers through an intuitive dashboard view with predictive and actionable insights, which means customers can eliminate guesswork from their recruitment marketing execution." Mishra concludes, "We utilize AWS for its scalability, reliability, and global presence, and we plan to continue using AWS as our preferred cloud technology platform."

About Joveo

Joveo, an AI-powered programmatic job advertising and candidate engagement platform, is revolutionizing talent sourcing and recruitment media buying for the world's largest and most intelligent employers, staffing businesses, RPOs, and media agencies. The Joveo platform helps businesses attract, source, engage, and hire the most relevant candidates on time and within their budget. Powering millions of jobs every day, the data-driven recruitment advertising platform employs advanced data science and machine learning to manage and optimize talent sourcing and applications across all online channels.

AWS Services Used



Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 700 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)

Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance. S3 is built to store and retrieve any amount of data from anywhere.

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AWS Batch

AWS Batch lets developers, scientists, and engineers efficiently run hundreds of thousands of batch and ML computing jobs while optimizing compute resources, so you can focus on analyzing results and solving problems.

[Learn more »](#)

Amazon EKS

Amazon Elastic Kubernetes Service (Amazon EKS) is a managed Kubernetes service to run Kubernetes in the AWS cloud and on-premises data centers.

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